# Devin Liang

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## EDUCATION

**UC Berkeley, Berkeley, CA, USA** Expected June 2028
B.A. Applied Mathematics

## Projects

**Quant Platform** | Python, Streamlit, pandas, NumPy, Plotly June 2026 - Present
Personal Project

- Developed a python based multi asset research and backtesting platform for evaluating custom quantitative factors and allow for analyzing portfolio performance base on technical indicators
- Building a modular framework to combine multiple factors, adjust position exposure dynamically and evaluate strategies using equity curves, returns, volatility and maximum drawdown.

**MPC Lab** | JavaScript August 2023 - April 2025
Group Project

- Contributed to an educational platform that dynamically generates customizable problems using randomized parameters rather than relying on a fixed question bank
- Designed and implemented the core logic for the dynamic question and solution generator for all subjects and the supporting functions for all educational tools
- Developed reusable helper functions throughout the project to reduce redundant computation and improve code efficiency

**Heat Equation Simulator** | Python, NumPy, Matplotlib, Tkinter April 2026 - May 2026
Personal Project

- Developed an interactive simulator for solving the 1D and 2D heat equations using explicit finite difference methods with numerical stability constraints based on spatial and temporal step sizes
- Used Matplotlib to animate temperature evolution for comparison between numerical and analytical solutions
- Designed an interactive GUI for configuring simulation parameters and switching between one and two dimensional models and support for importing custom initial conditions from CSV/TXT files

## Experience

**Math Tutor** April 2024 - August 2026
Math Clinic | Los Angeles, CA

- Maintained weekly performance records for 20 students and adjusted instruction based on individual progress and recurring learning gaps
- Coached a student to 2nd place in a county level mathematics competition

**Hackathon Participant** January 2025
Irvine Hacks 2025 | Irvine, CA

- Integrated Google Maps Geocoding and JavaScript Maps APIs to retrieve location coordinates and identify nearby established shelters
- Implemented proximity ranking logic to identify and prioritize the nearest shelters

## Leadership

**Boy Scouts of America** June 2019 - December 2023
Assistant Senior Patrol Leader | Los Angeles, CA

- Coordinated weekly trainings for 8 patrol leaders to help with guiding their individual patrol groups
- Support patrol leaders for planning and organizing meetings and events for 20+ scouts
- Managed a 30+ person church garden restoration project by coordinating schedules and responsibilities while tracking project expenses for materials, equipments and foods

## SKILLS

- **Programming Languages:** Python, C++, Java, JavaScript, HTML/CSS
- **Libraries:** NumPy, pandas, Matplotlib, Plotly, Streamlit, Tkinter
- **Tools:** Git, GitHub, MATLAB, SolidWorks, Excel
- **Language:** English (fluent); Mandarin (native); Cantonese (native); Japanese (basic)